

Lecture 5

ChE 467/596-012 Polymer Rheology

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Last Class & Moving Forward

Last Class ended with:

- **Dynamic oscillatory shear**

Today's Agenda

- **Start-up of steady shear – stress growth**
- **Stress relaxation following steady shear**
- **Step strain -stress relaxation following sudden strain**
- **Creep and recovery**

Material Functions for dynamic oscillatory experiments

G' = elastic modulus $\longrightarrow \eta'' \equiv \text{Dynamic Rigidity} = \frac{G'}{\omega}$

G'' = viscous modulus $\longrightarrow \eta' \equiv \text{Dynamic Viscosity} = \frac{G''}{\omega}$

$$\begin{aligned}\tau_{yx} &= G' \gamma_o \sin(\omega t) + G'' \gamma_o \cos(\omega t) \\ &= \eta''(\omega) \dot{\gamma}_o \sin(\omega t) + \eta'(\omega) \dot{\gamma}_o \cos(\omega t)\end{aligned}$$

$$\tan \delta \equiv \text{loss tangent} = \frac{G''}{G'}$$

$\eta^* \equiv \text{Complex Viscosity}$

$$= \left[\eta'^2 + \eta''^2 \right]^{1/2} = \left[\left(\frac{G''}{\omega} \right)^2 + \left(\frac{G'}{\omega} \right)^2 \right]^{1/2}$$

Significance of Material Functions

G' =elastic modulus

Energy stored in system— shape and magnitude indicates whether sample an elastic network, gel, liquid

G'' =viscous modulus

Energy dissipated – relative magnitude w.r.t. G' indicates state of material also

$$\tan \delta \equiv \text{loss tangent} = \frac{G''}{G'}$$

Relative importance of viscous nature over elastic nature; usually <1 gels, solids (one criteria); usually >1 viscous fluids

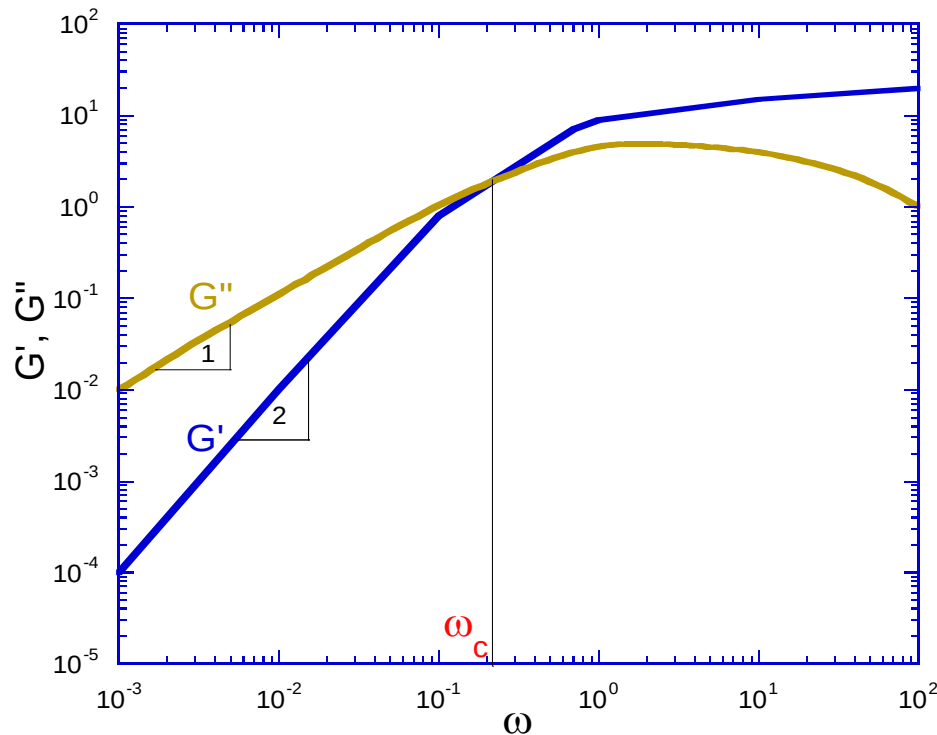
η^* $\equiv$ Complex Viscosity

$$= \left[\eta'^2 + \eta''^2 \right]^{1/2} = \left[\left(\frac{G''}{\omega} \right)^2 + \left(\frac{G'}{\omega} \right)^2 \right]^{1/2}$$

Analogous to steady viscosity for conventional polymeric system

Material Functions, cont'd

Typical polymer melt dynamic data



$$\omega_c = \dot{\gamma}_c$$

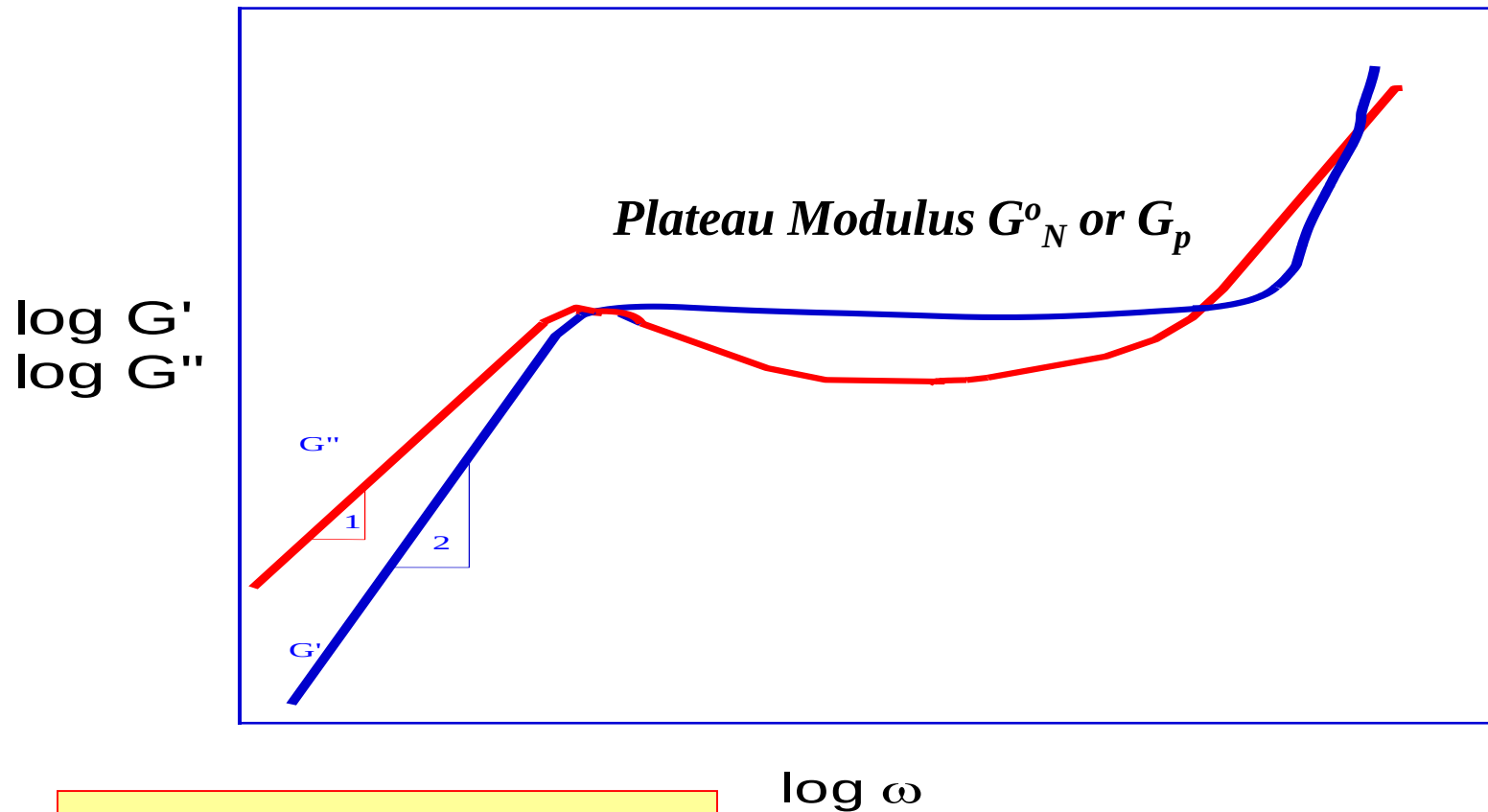
$$1/\omega_c = 1/\dot{\gamma}_c = \lambda$$

λ – longest relaxation time

$$G' \sim \omega^2 ; G'' \sim \omega \text{ at low } \omega$$

Material Functions, cont'd

Typical polymer melt dynamic data



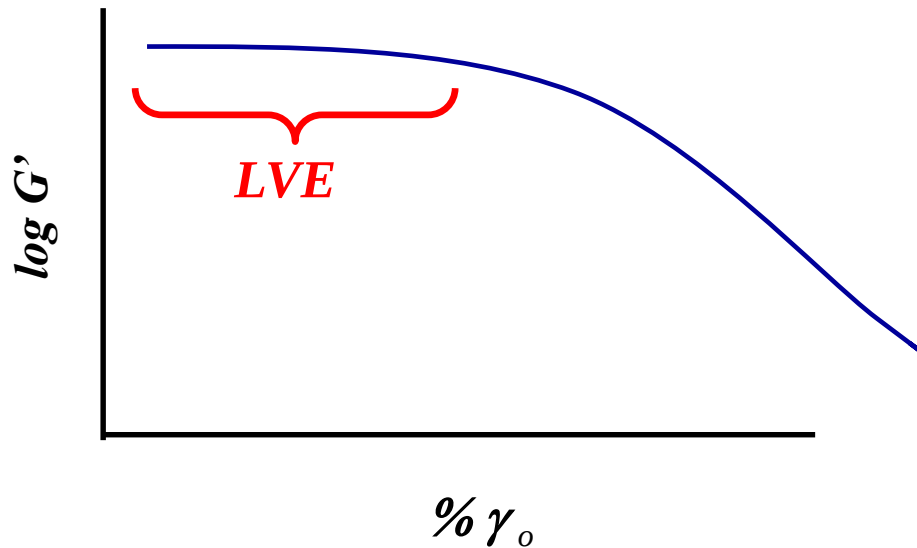
$$G' \sim \omega^2 ; G'' \sim \omega \text{ at low } \omega$$

Elastic and Viscous Moduli (G' , G'')

- The frequency sweep experiment was done at a constant strain amplitude, γ_o
- What strain amplitude do we use?

Material Functions, cont'd

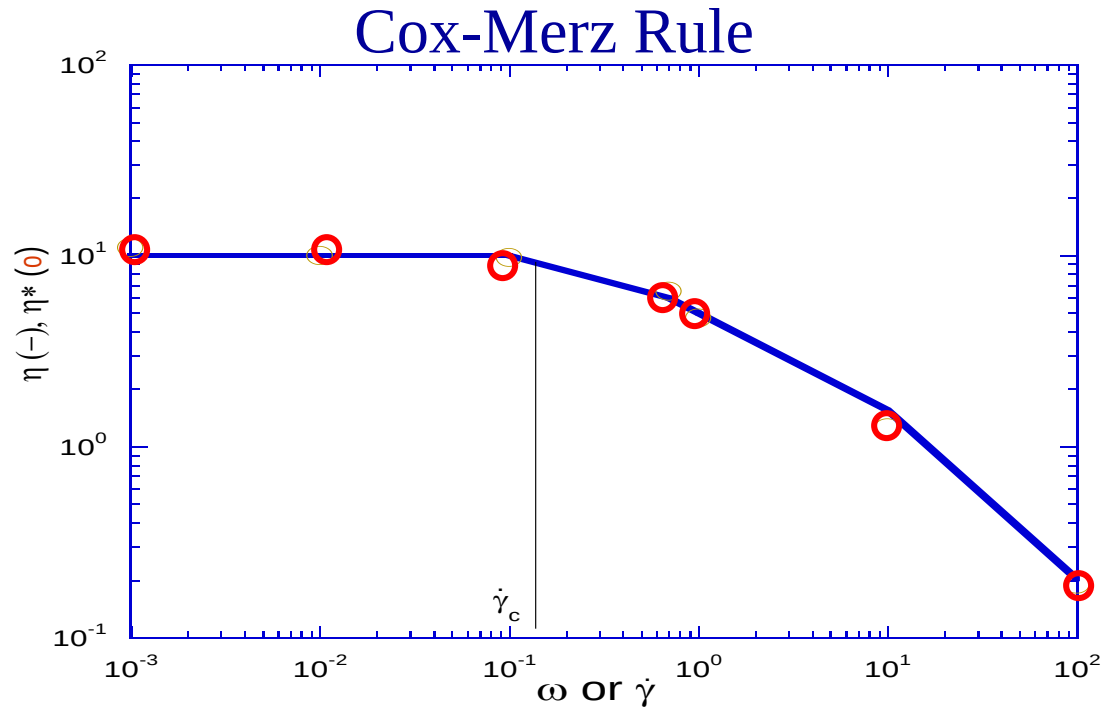
- All dynamic material functions discussed represent linear viscoelastic properties



**LVE = linear
viscoelastic
regime**

- Experiments have to be done at v. small strain amplitude
- Do strain sweep first to get LVE; then do frequency sweep at a strain in the LVE regime

Material Functions, cont'd



$$\eta^*(\omega) = \eta(\dot{\gamma}) \quad @ \quad \omega = \dot{\gamma}$$

$$\omega_c = \dot{\gamma}_c; \quad \frac{1}{\omega_c} \sim \frac{1}{\dot{\gamma}_c} = \lambda \quad \lambda \equiv \text{longest relaxation time}$$

In-class exercise

- Plot η' and η'' vs ω for a polymer

$$\eta'' \equiv \frac{G'}{\omega}$$

$$\eta' \equiv \frac{G''}{\omega}$$

- Are there any instances that you would prefer doing dynamic experiments over steady shear?

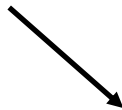
G^* in terms of Viscous and Elastic Components

$$G^* = \frac{\tau_0}{\gamma_0} \quad G^* \equiv \text{amplitude ratio}$$



The *loss modulus* is given by the ratio of the stress amplitude to the strain amplitude times the sine of the phase shift angle

$$|G^*| = \sqrt{(G')^2 + (G'')^2} \longrightarrow G'' = \frac{\tau_0}{\gamma_0} \sin \delta$$



$$G' = \frac{\tau_0}{\gamma_0} \cos \delta$$

The *shear storage modulus* is given by the ratio of the stress amplitude to the strain amplitude times the cosine of the phase shift.

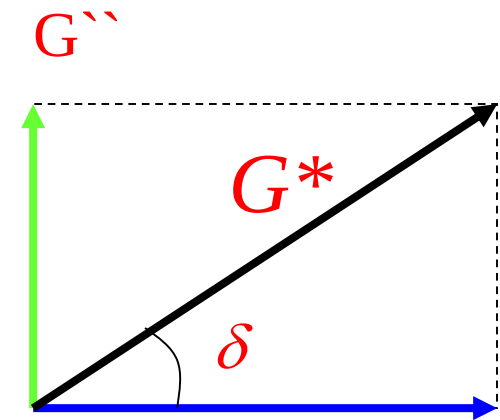
G* in terms of complex notation

$$G^* = \frac{\tau_0}{\gamma_0} \quad G^* \equiv \text{amplitude ratio}$$

↓ Expressed in complex notation

$$G^* = G' + iG''$$





$$G'' = \frac{\tau_0}{\gamma_0} \sin \delta$$



$$G' = \frac{\tau_0}{\gamma_0} \cos \delta$$

The *shear storage modulus* is the “real” component of G^*

The *shear loss modulus* is the “imaginary” component of G^*

Rest of Today's Agenda

Time-dependent/transient shear experiments

start-up of shear flow

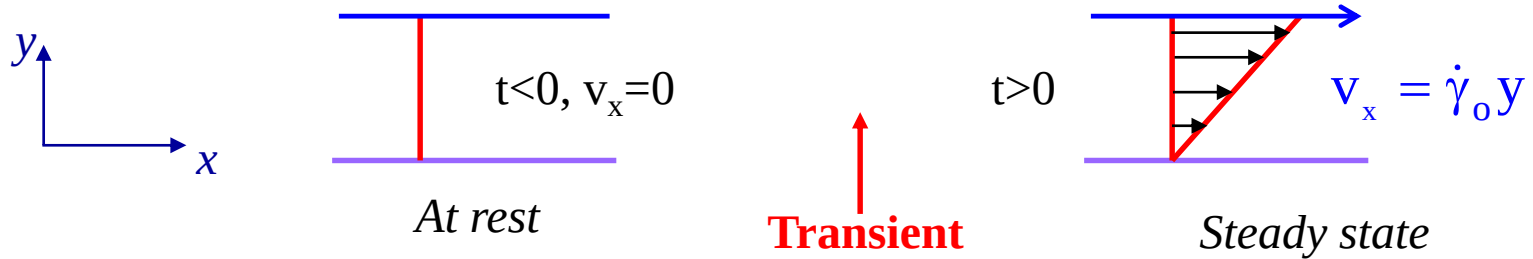
- look at initial transient conditions before flow reaches steady state

stress relaxation following deformation

- relaxation of stress upon cessation of steady shear flow
- stress relaxation following step strain

Creep experiments

Stress Growth (Start Up)



$$\tau_{yx} = \eta^+(t, \dot{\gamma}_o) \dot{\gamma}_o$$

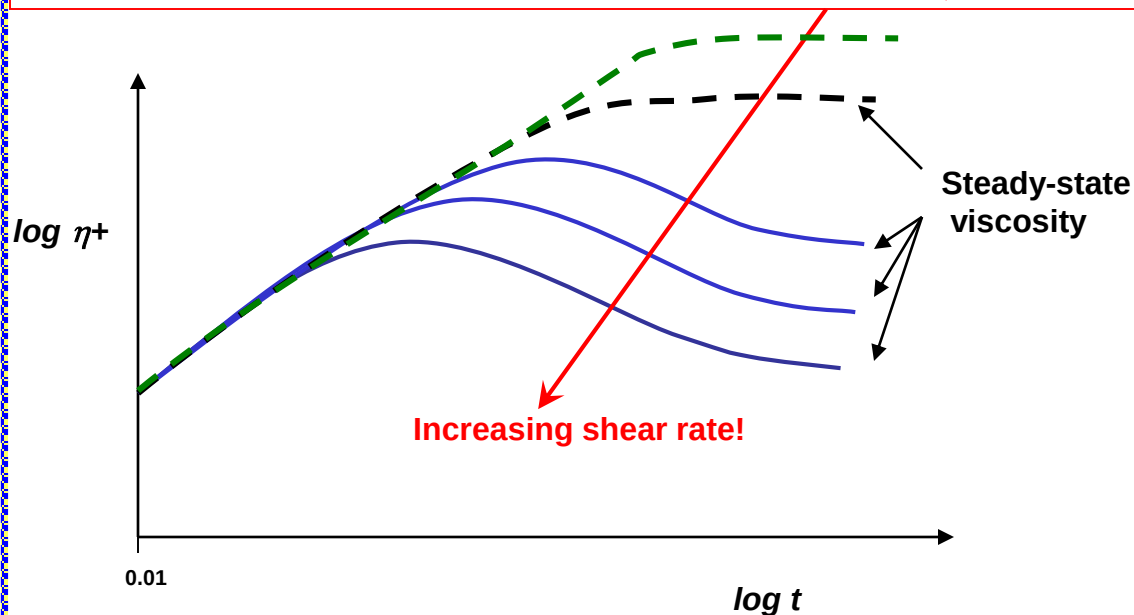
$$\tau_{xx} - \tau_{yy} = \psi_1^+(t, \dot{\gamma}_o) \dot{\gamma}_o^2$$

$$\tau_{yy} - \tau_{zz} = \psi_2^+(t, \dot{\gamma}_o) \dot{\gamma}_o^2$$

↑
material functions

- 3 transient material functions
- “+” emphasizes steady rate applied for positive times

Stress Growth (Start Up)



- Low shear rate -viscosity/stress increase monotonically with time
- Increasing shear rate - deviation from envelope
- Higher shear rates - observe stress/viscosity overshoot
- Envelope corresponds to linear regime

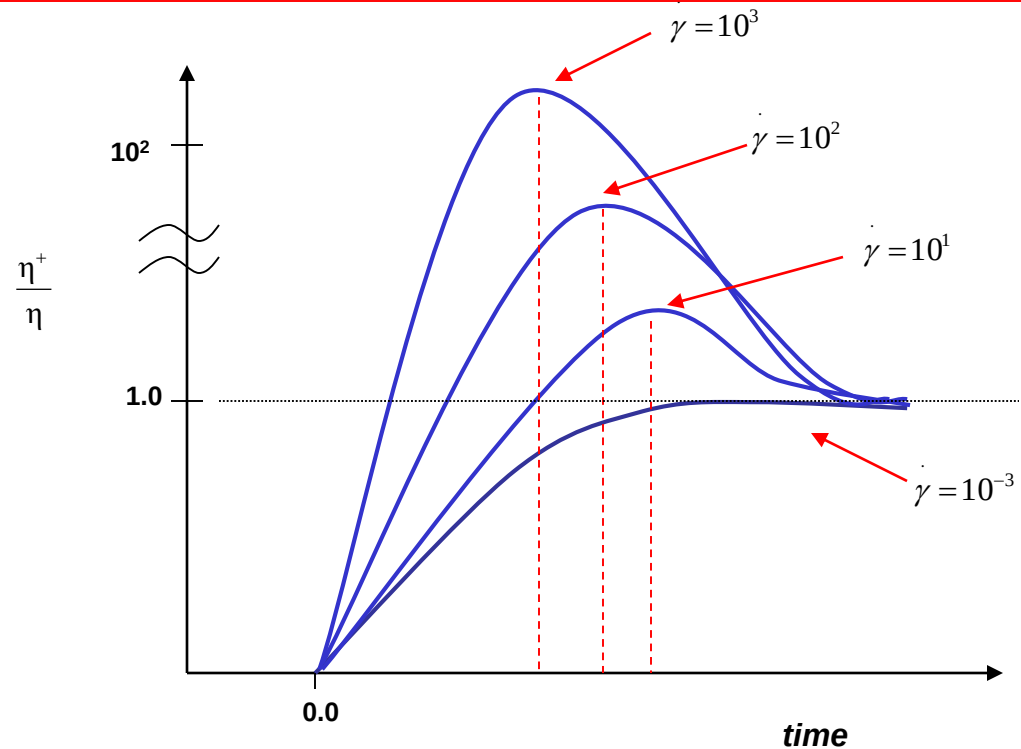
Stress Growth (cont'd)

• *What can we say?*

• **Relative overshoot increases with increasing shear rate!**

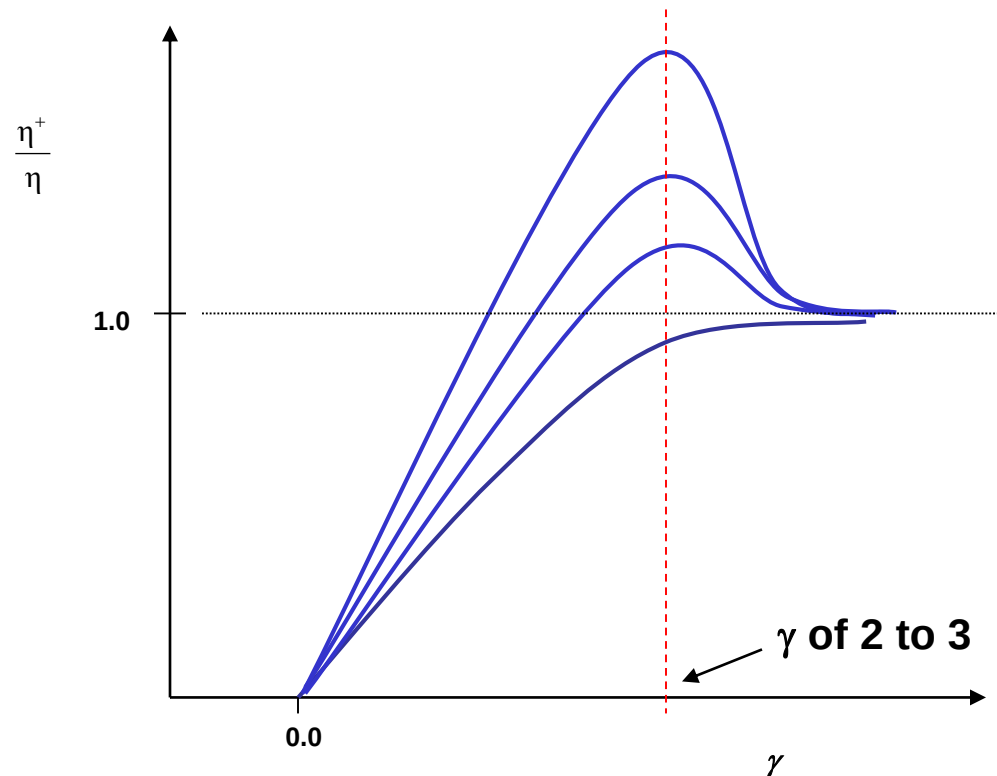
• **Maxima shifts to shorter time with increasing shear rate**

• *Any ramifications of stress overshoot?*



Stress Growth (cont'd)

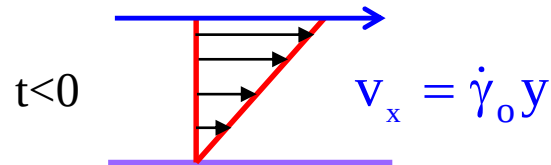
- Overshoot occurs at same strain level although different times
- strain level $\sim 2-3$
- Note $\gamma = \dot{\gamma}_0 t$



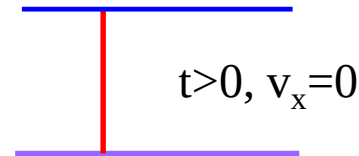
Similar behavior observed in normal stress with overshoot at $\gamma \sim 2$

Stress Relaxation

Stress relaxation following steady shear



Steady state



At rest

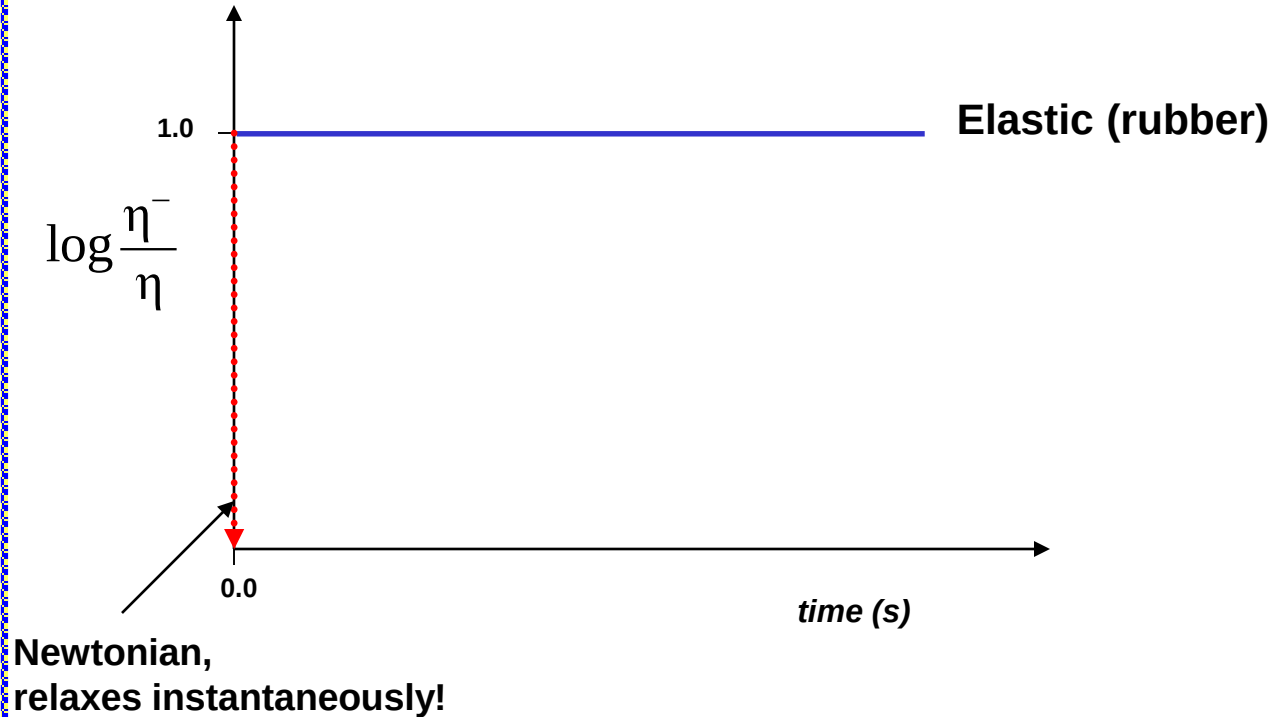
$$\tau_{yx} = \eta^-(t, \dot{\gamma}_0) \dot{\gamma}_0$$

$$\tau_{xx} - \tau_{yy} = \psi_1^-(t, \dot{\gamma}_0) \dot{\gamma}_0^2$$

$$\tau_{yy} - \tau_{zz} = \psi_2^-(t, \dot{\gamma}_0) \dot{\gamma}_0^2$$

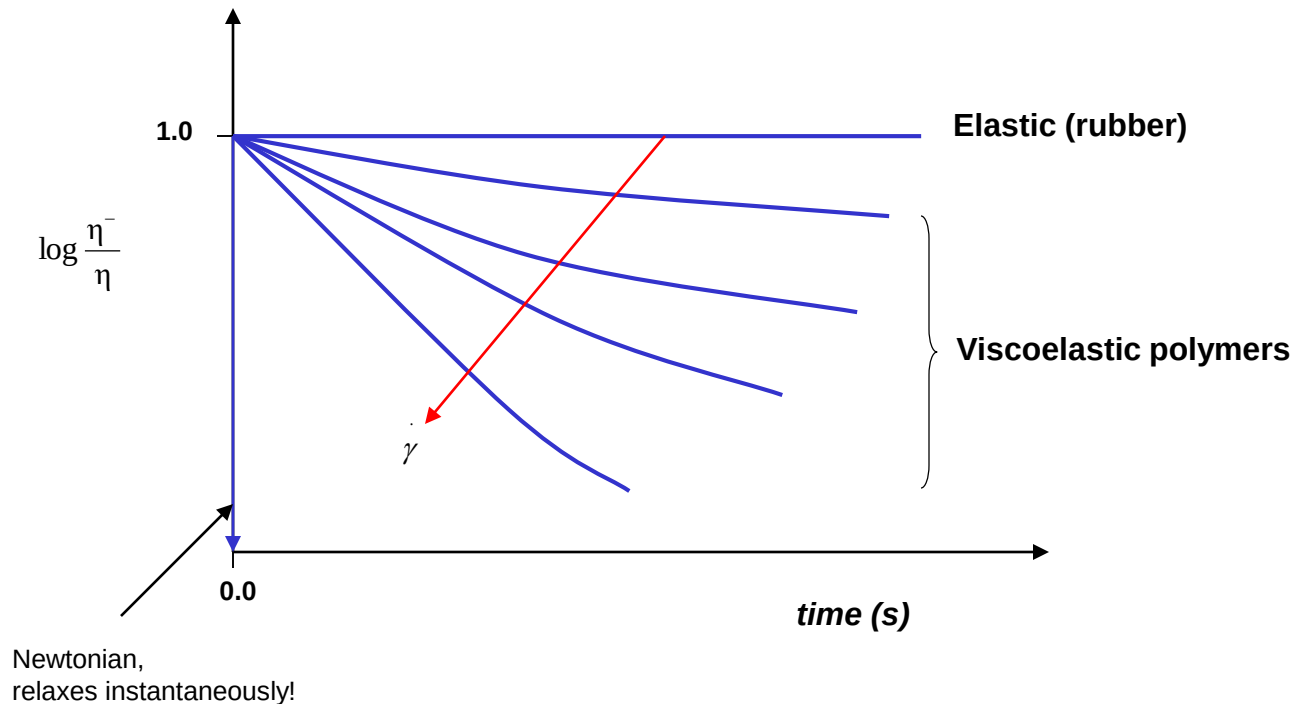
- 3 more transient material functions
- “-” emphasizes steady rate applied for previous times

Stress relaxation – after steady shear



How do an elastic material or Newtonian fluid behave?

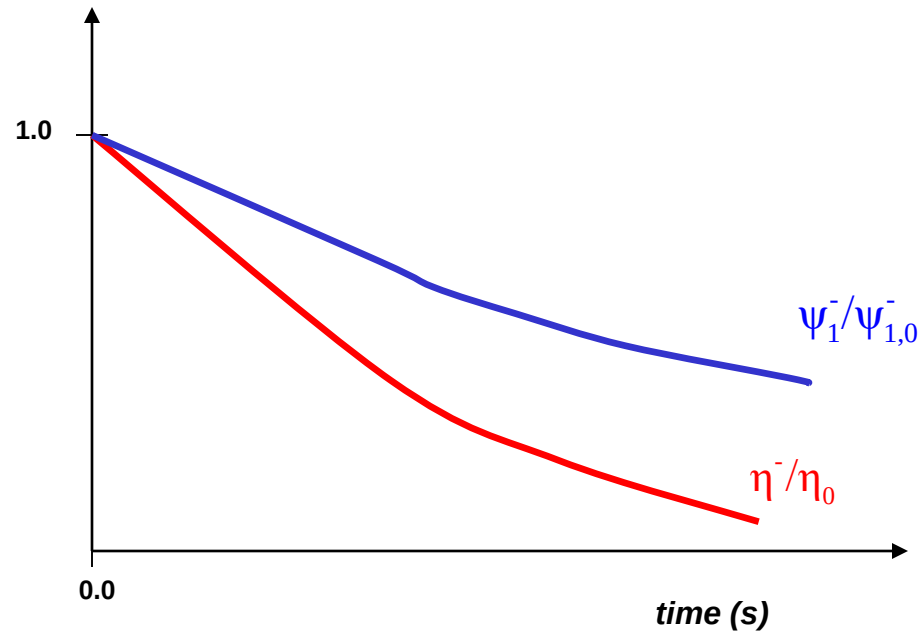
Stress relaxation – after steady shear



- Increasing initial shear rate – relaxation faster
- Stresses eventually relax to zero (not shown in drawing)

Stress relaxation – after steady shear

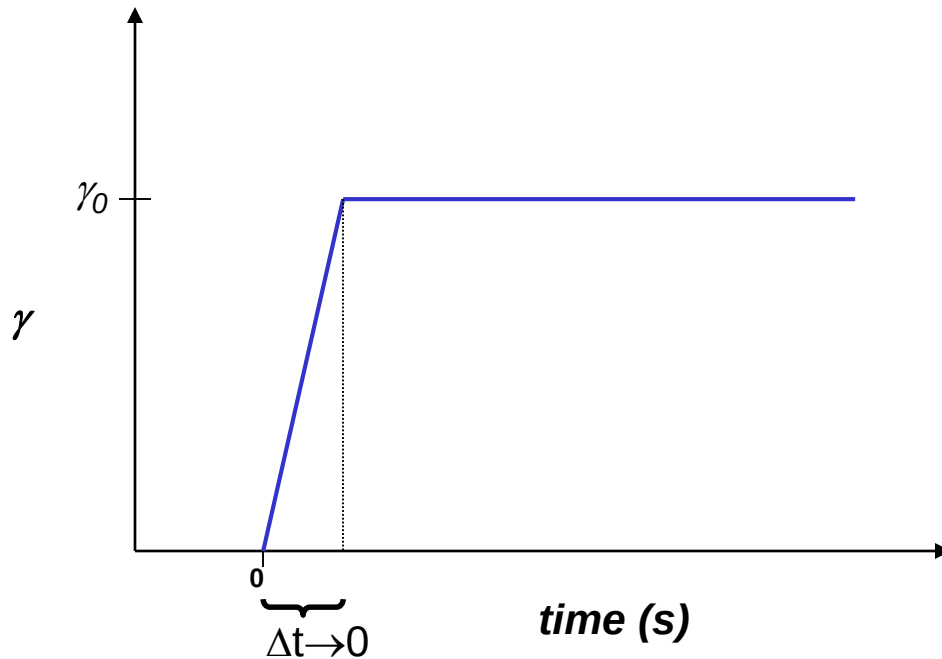
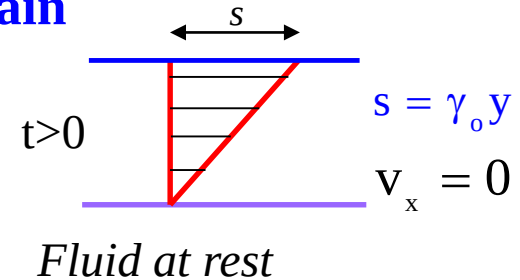
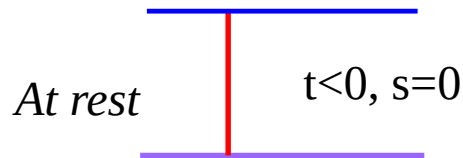
Normal vs shear stresses



- *Normal stresses relax more slowly*
- *Any implications?*

Stress relaxation

Stress relaxation following step shear or strain

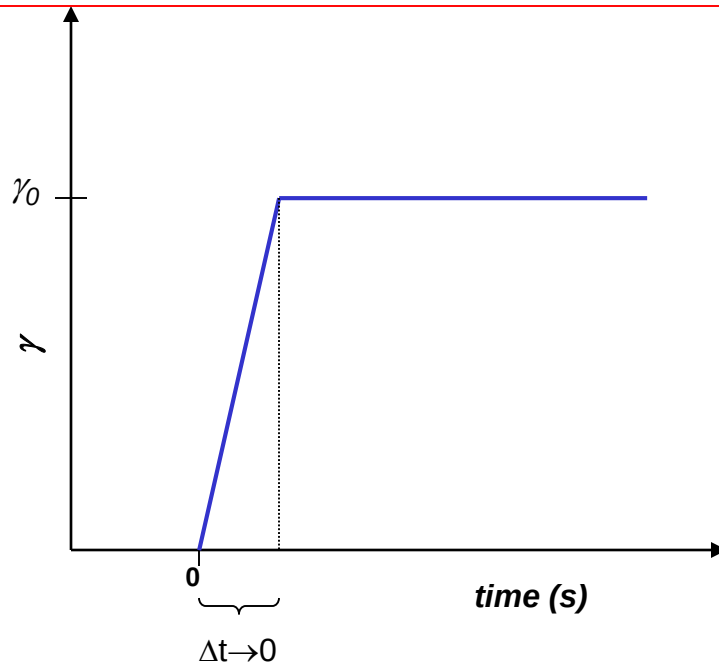


Apply strain
instantaneously

$$\Delta t \rightarrow 0$$

$$\gamma_o = \dot{\gamma} \Delta t$$

Stress relaxation-step strain

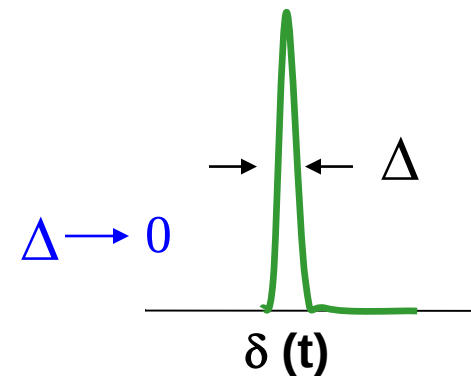


$$\gamma_o = \dot{\gamma} \Delta t$$

What is the shear rate?

$$\dot{\gamma} = \dot{\gamma}_o \delta(t)$$

Dirac delta function

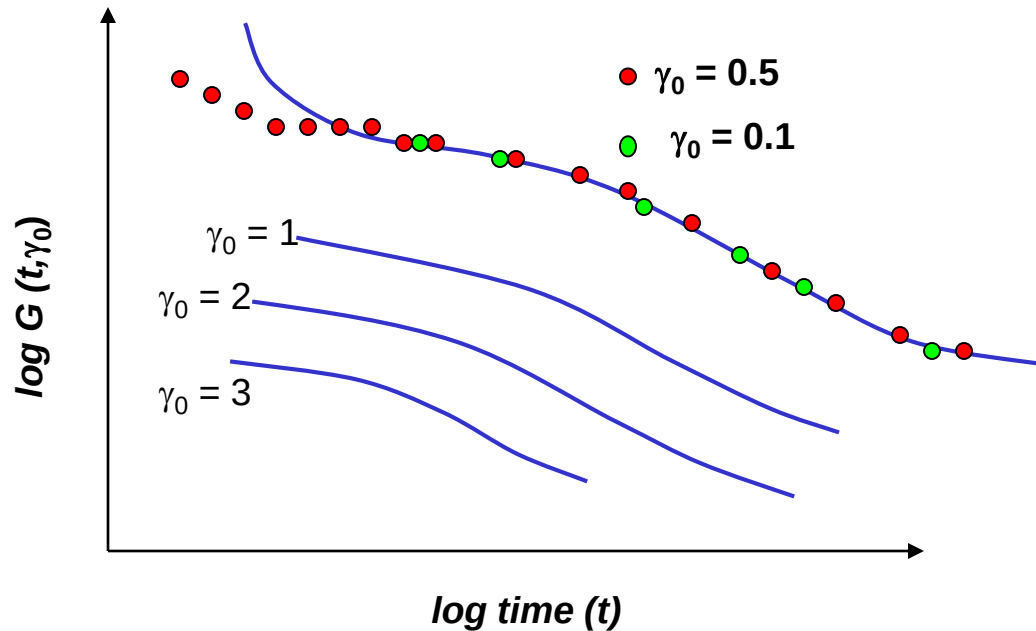


Stress-strain relation

$$\tau_{yx} = G(t, \gamma_o) \gamma_o \longrightarrow \tau_{yx} = G(t) \gamma_o \text{ in the linear viscoelastic regime}$$

G(t) – relaxation modulus

Stress relaxation-step strain

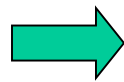


There is overlap for a certain strain regime

⇒ This is the LVE, or linear, regime! The relaxation is independent of strain level

Increasing strain shifts relaxation curve down but no change in shape

⇒ There is time strain separability

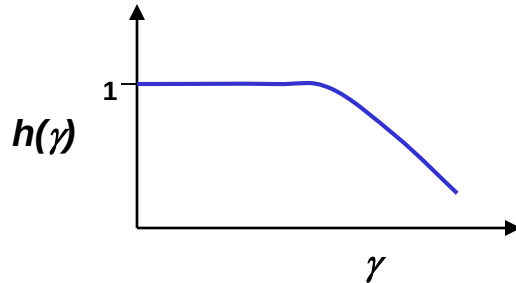


$$G(t, \gamma) = G(t) h(\gamma)$$

$h(\gamma)$ is the damping function

Stress relaxation-step strain

Damping function

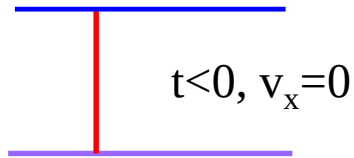


$h(\gamma)$ is a measure of non-linearity

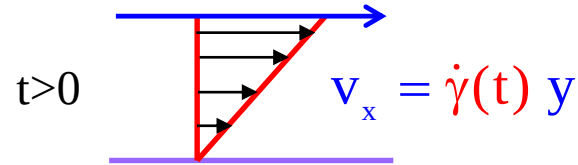
Questions

- What is $h(\gamma)$ in the LVE?
- What is $h(\gamma)$ for an elastic material?
- Plot $G(t)$ for an elastic solid & Newtonian fluid

Creep Experiment



At rest



Apply constant stress τ_o

Possible Experimental sequence

Apply constant stress τ_o

Measure γ or creep compliance $J(t, \tau_o)$

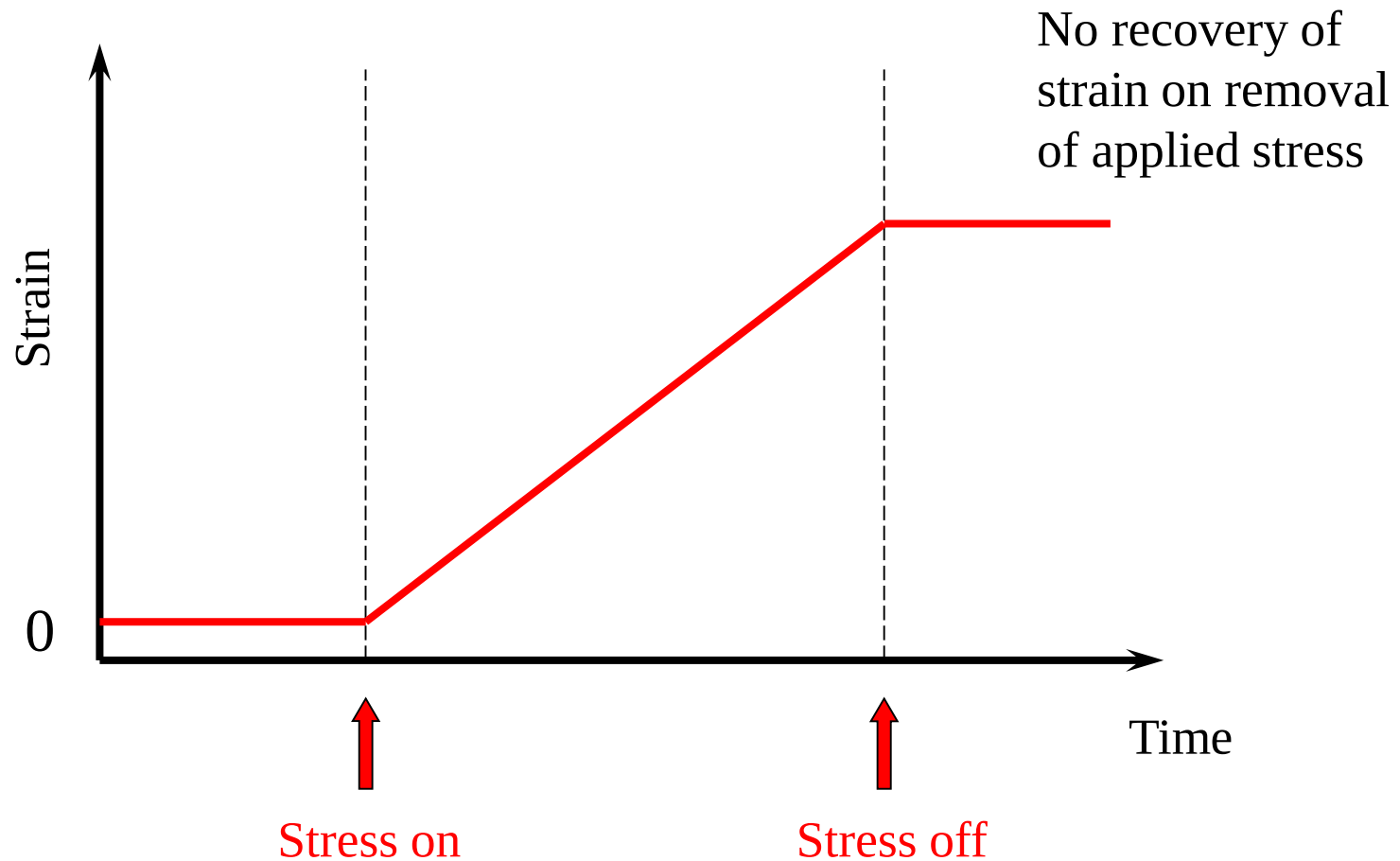
$$J(t, \tau_o) = \gamma(t) / \tau_o$$

Remove stress τ_o , i.e. set $\tau_o = 0$

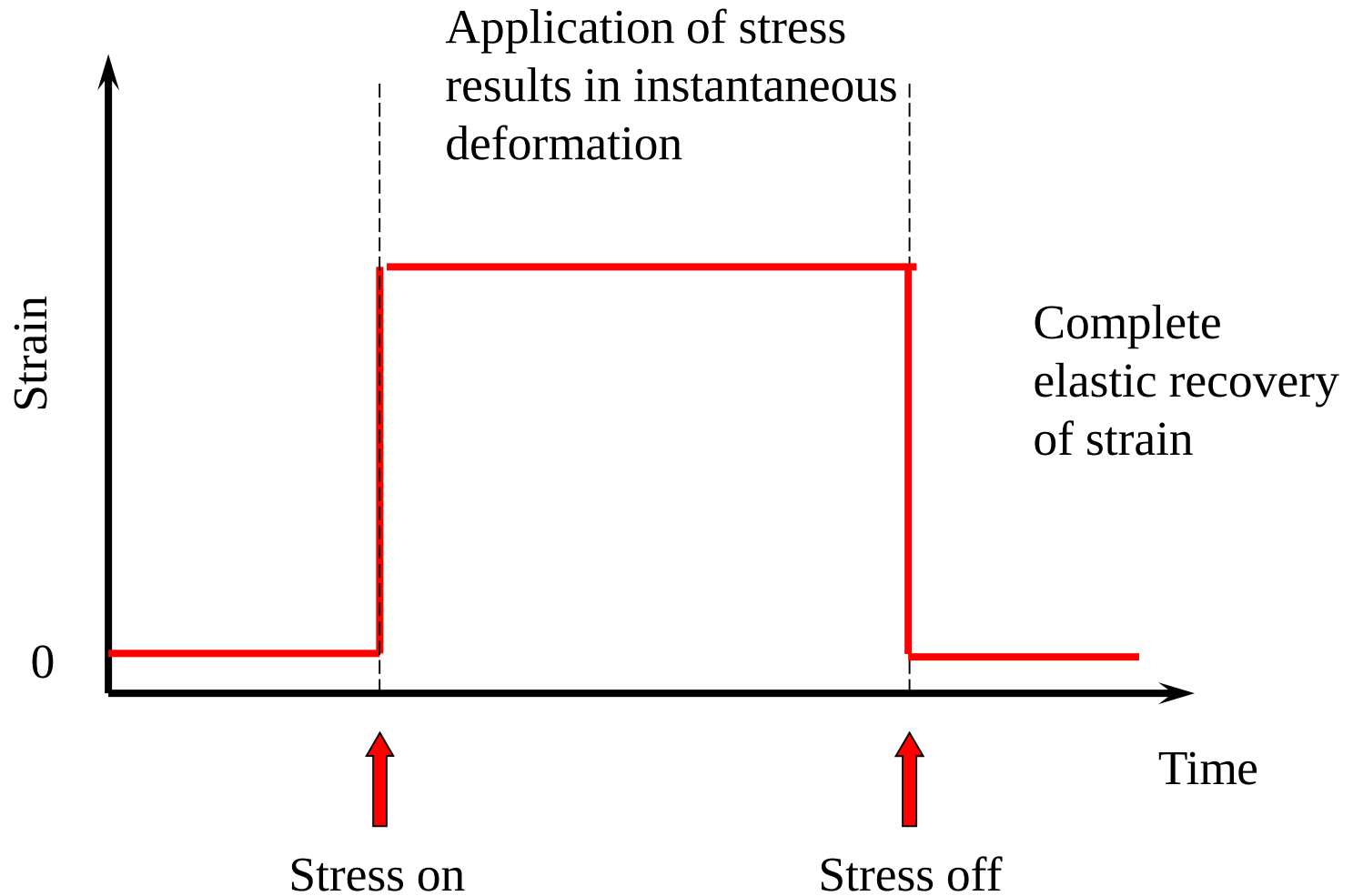
**Measure recoverable strain γ_r
or creep recovery J_r**

Note that recovery is also referred to as recoil

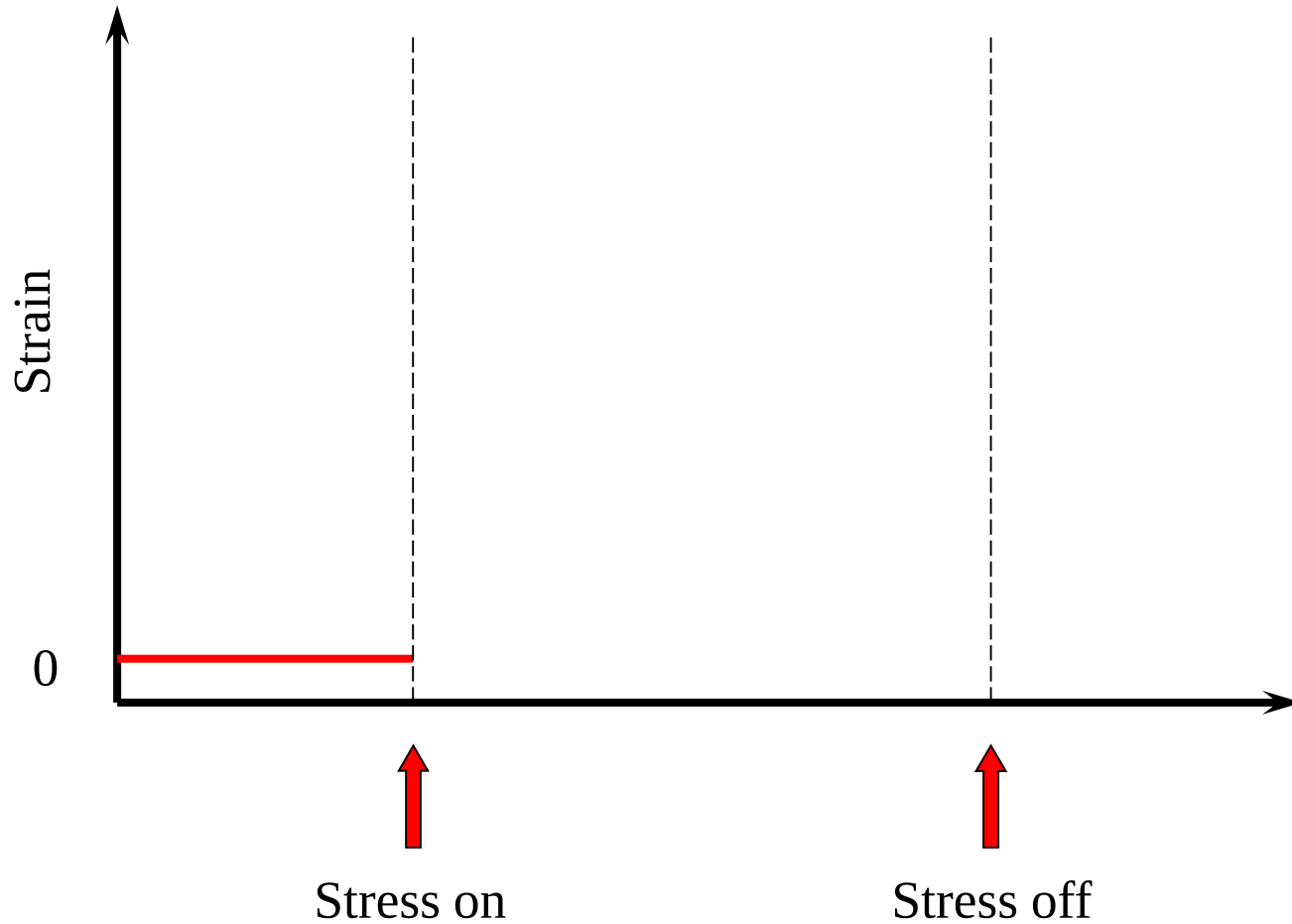
Strain - Time Plot : Perfect Liquid



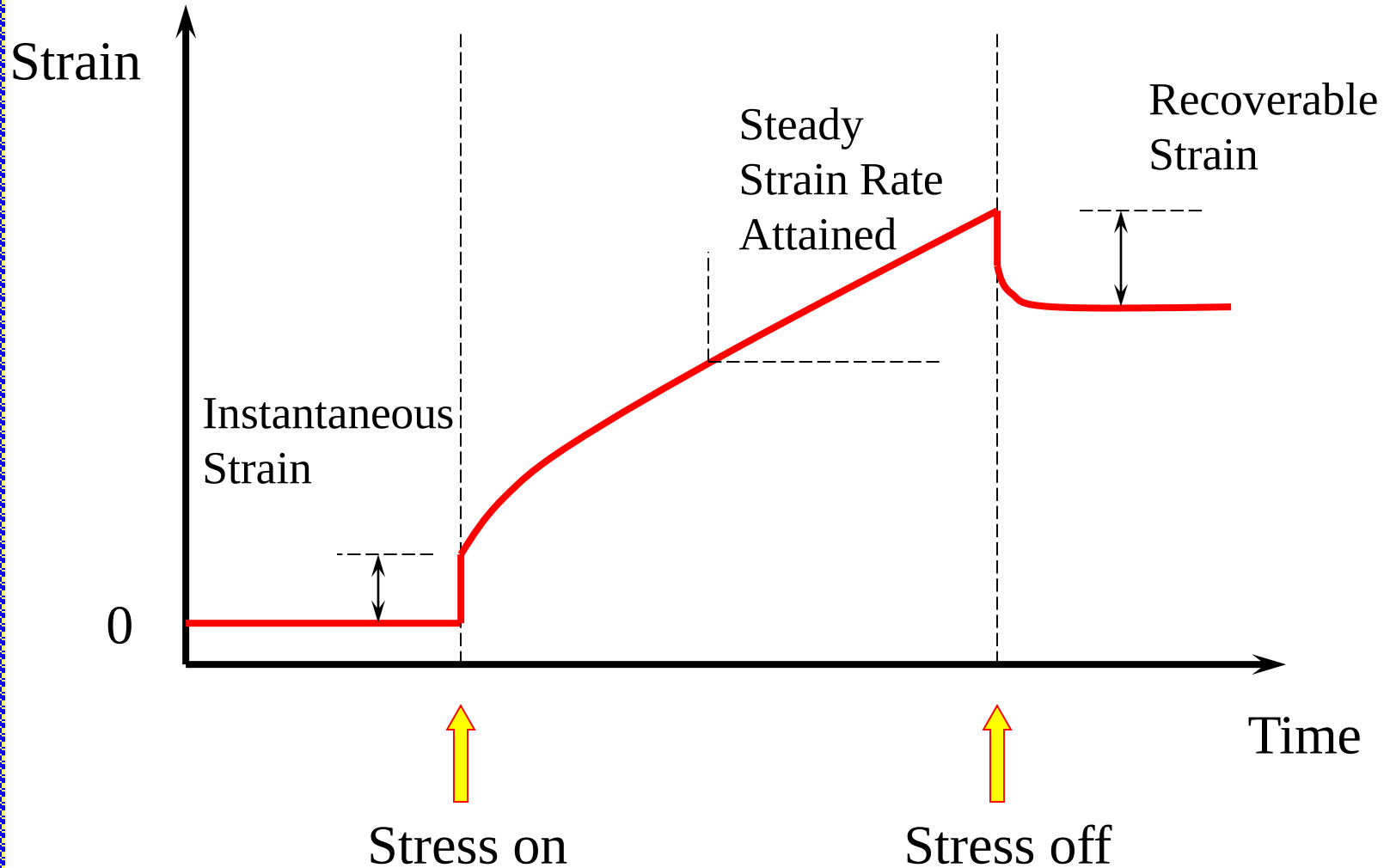
Strain - Time Plot : Elastic Mat'l



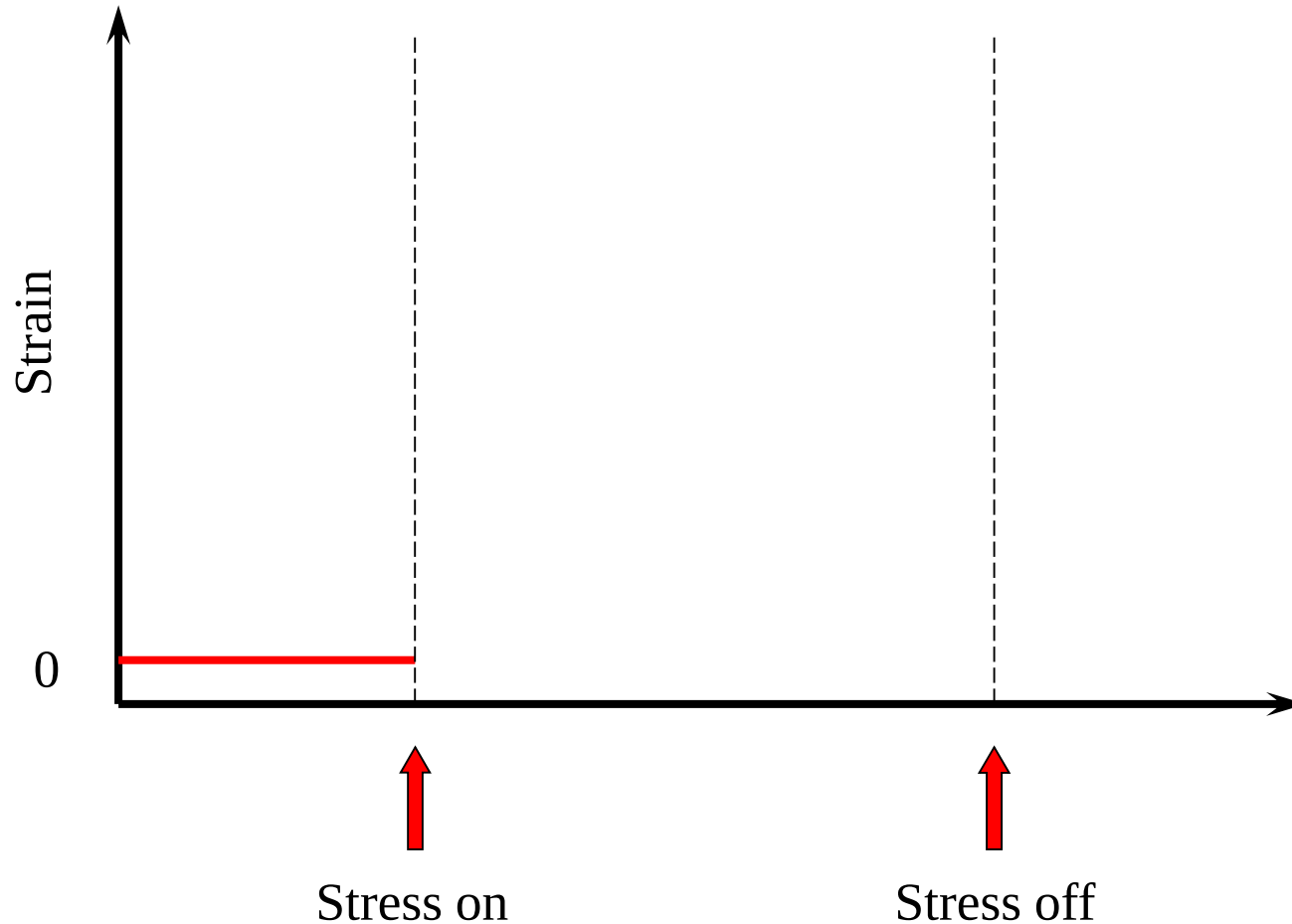
Strain - Time Plot : Viscoelastic “Fluid”



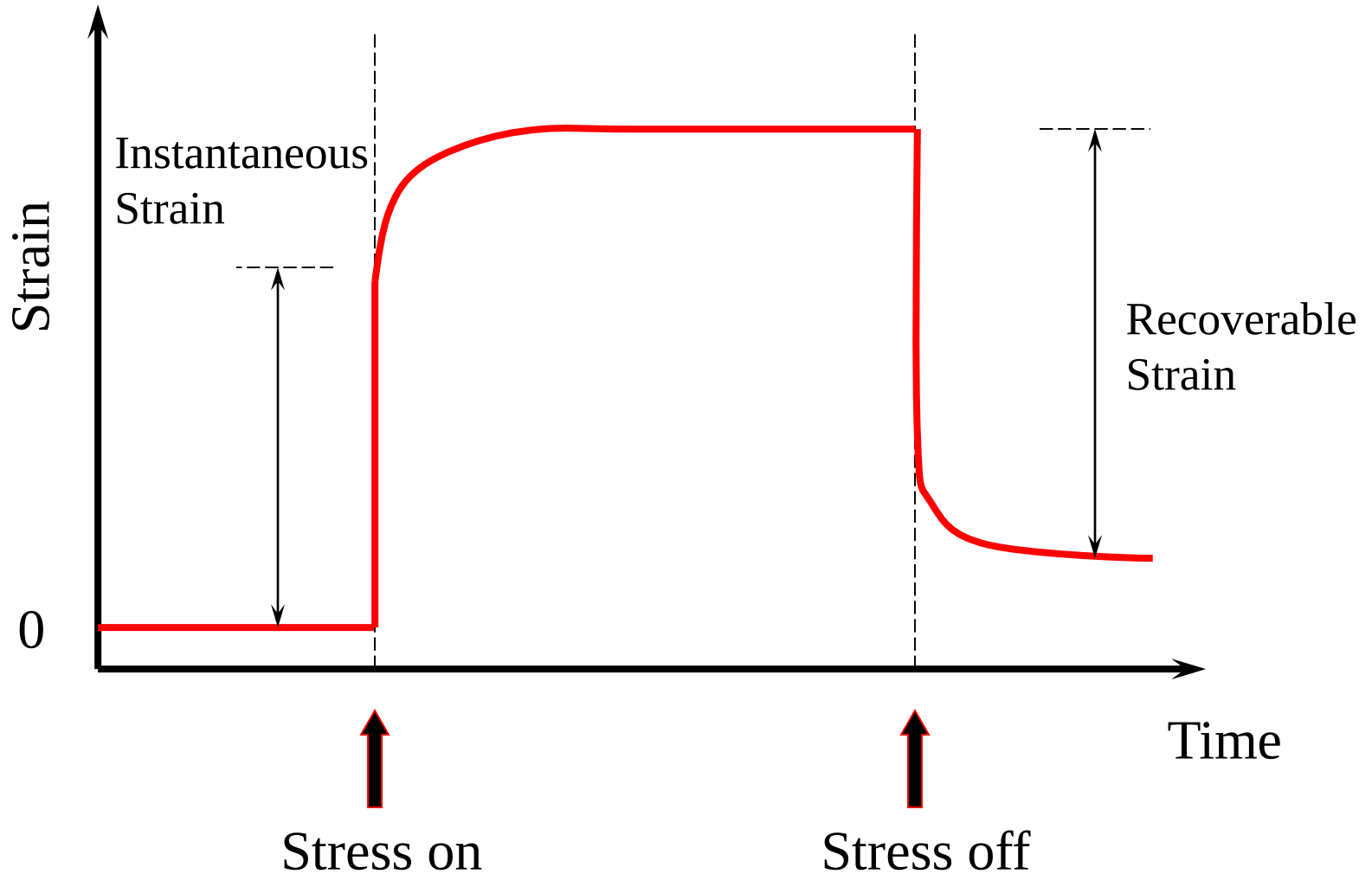
Strain - Time Plot : Viscoelastic “Fluid”



Strain - Time Plot : Viscoelastic “Solid”



Strain - Time Plot : Viscoelastic “Solid”



Recap- Possible experiments

- **Steady shear**
- **Dynamic oscillatory shear**
- **Start-up of steady shear – stress growth**
- **Stress relaxation following steady shear**
- **Step strain -stress relaxation following sudden strain**
- **Creep and recovery**